MALLA REDDY TECHNICAL CAMPUS

A Constituent Unit of MALLA REDDY
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SUBJECT:- DESIGN AND ANALYSIS OF ALGORITHMS

TOPIC:- AGRICULTURAL IRRIGATION SCHEDULE



Smart Agriculture Irrigation Schedule

A simple algorithm decides

WHEN • HOW MUCH • WHICH FIELD

water should be supplied using soil, crop and weather information.

- Reduce unnecessary watering.
- Keep soil moisture near the target range.
- Create a repeatable daily/weekly plan.

Problem

Manual watering may waste water or miss the crop's ideal irrigation time.

Idea

Convert field data into a clear schedule using simple rules and an algorithm.



Inputs & Decision Factors

Soil Moisture

Current moisture % from sensor.

Crop Stage

Seedling, growth, flowering or harvest.

Weather

Rain forecast, temperature and humidity.

Field Area

Area and crop water requirement.

Water Level

Available tank / reservoir capacity.

Time

Previous irrigation and next allowed slot.

Key rule: use fresh sensor/weather data before generating the next schedule.



Irrigation Decision Logic

1. **READ** — Get soil moisture, crop stage, forecast and water availability.
2. **CHECK** — Is soil moisture below the crop's lower limit?
3. **RAIN TEST** — If useful rain is expected soon, postpone or reduce watering.
4. **CALCULATE** — Estimate the required water amount from field + crop need.
5. **SCHEDULE** — Choose the safest time slot and assign duration/volume.



Algorithm — Easy Pseudocode

```
START
  Read moisture, crop stage, rain forecast and tank level
  IF moisture < lower_limit
      IF rain_expected = YES
           postpone / reduce irrigation
       ELSE
          calculate water_need
           create irrigation slot
       END IF
  ELSE
      skip irrigation and monitor
  END IF
  Save today's schedule
STOP
```

Simple logic • easy to implement • easy to explain



Example 7-Day Smart Schedule

Day	Soil	Rain	Decision	Action
Mon	Dry	No	**IRRIGATE**	18 min • Zone A
Tue	Good	Yes	**WAIT**	No water
Wed	Good	No	**MONITOR**	Sensor check
Thu	Dry	No	**IRRIGATE**	20 min • Zone A
Fri	Good	No	**WAIT**	No water
Sat	Low	No	**IRRIGATE**	15 min • Zone B
Sun	Good	Yes	**WAIT**	Rain used

Example only: real schedules should use the actual crop, soil, weather and irrigation-system data.



Benefits, Limits & Complexity

Benefits

- Saves water
- Reduces over-irrigation
- Uses data, not guesswork
- Creates consistent schedules

Limits

- Sensors need calibration
- Weather forecasts can change
- Crop limits vary
- Pump flow must be known

DAA View

- Each field can be processed in one pass
- Basic decision logic is O(n)
- Scheduling can be extended with priority rules

Future upgrades

IoT sensors → weather API → prediction → automatic valves → continuous learning



THANK YOU

Smart Agriculture Irrigation Schedule

Design and Analysis of Algorithms

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